Week-1 Assignment (UAPOML)

1) (a)

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

1st month

$$= \frac{108-100}{100} = \frac{8}{100} \Rightarrow 8\%$$

2nd month $= \frac{103-108}{108} = \frac{-5}{108} \Rightarrow -4.63\%$

3rd month $= \frac{115-103}{103} = \frac{12}{103} \Rightarrow 11.65\%$

4th month $= \frac{110-115}{115} = \frac{-5}{115} \Rightarrow -4.34\%$

(b)

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

1st month $= \ln\left(\frac{108}{100}\right) = 7.69\%$

2nd month $= \ln\left(\frac{103}{108}\right) = -4.72\%$

3rd month $= \ln\left(\frac{115}{103}\right) = 11.02\%$

4th month $= \ln\left(\frac{110}{115}\right) = -4.44\%$

(c) $|R_t - r_t|$ for 1^{st} month $= |8 - 7.69| = 0.3\%$

2^{nd} month $= |-4.63 - (-4.72)| = 0.09\%$

3^{rd} month $= |11.65 - 11.02| = 0.62\%$

4^{th} month $= |-4.34 - (-4.44)| = 0.09\%$

for 3rd month it's least accurate because 11.65 was the largest of all simple returns (magnitude)
The approximation used here is $\ln(1+x) \approx x$ whose error grows as $\frac{x^2}{2}$.

2) (a) linearity of expectation is

$$E(aX + bY) = aE(X) + bE(Y)$$

here,

$$R_p = \omega_1 R_1 + \omega_2 R_2$$

$$E[R_p] = \omega_1 E[R_1] + \omega_2 E[R_2]$$

$$= 0.4 \times 15\% + 0.6 \times 7\% = 6\% + 4.2\%$$

$$= 10.2\%$$

(b) As we can see in the relation there is no term of covariance or any related term.
This is regardless of dependence.
So, the result will not change.

3)

$$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$$

$$R_p = \omega_1 R_1 + \omega_2 R_2$$

wkt,

$$Var(aX) = a^2 Var(X) \text{ and } Cov(aX, bY) = ab\,Cov(X,Y)$$

So,

$$Var(R_p) = Var(\omega_1 R_1) + Var(\omega_2 R_2) + 2Cov(\omega_1 R_1, \omega_2 R_2)$$

$$= \omega_1^2 Var(R_1) + \omega_2^2 Var(R_2) + 2\omega_1\omega_2 Cov(R_1, R_2)$$

multiplying $\sigma_1 \sigma_2$ to last term and dividing

$$= \omega_1^2\sigma_1^2 + \omega_2^2\sigma_2^2 + 2\omega_1\omega_2 \boxed{\frac{Cov(R_1,R_2)}{\sigma_1\sigma_2}} \times \sigma_1\sigma_2$$

$\hookrightarrow \rho_{12}$
$\hookrightarrow$ Correlation coefficient

$$\sigma_p^2 = \omega_1^2\sigma_1^2 + \omega_2^2\sigma_2^2 + 2\omega_1\omega_2\rho_{12}\sigma_1\sigma_2$$

4) (a) $\sigma_p^2 = (0.5)^2(25)^2 + (0.5)^2(12)^2 + 2(0.5)(0.5)\rho(25)(12)$
$= 192.25 + 150\rho$

for $\rho = +1$
$\sigma_p^2 = 192.25 + 150 = 342.25$
$\sigma_p = 18.5\%$

$\rho = 0$
$\sigma_p^2 = 192.25$
$\sigma_p = 13.87\%$

$\rho = -1$
$\sigma_p^2 = 192.25 - 150$
$\sigma_p^2 = 42.25$
$\sigma_p = 6.5\%$

(b) σ_p is same as weighted average when $\rho = 1$
putting $\rho = 1$
$\sigma_p^2 = \omega_1^2\sigma_1^2 + \omega_2^2\sigma_2^2 + 2\sigma_1\sigma_2\omega_1\omega_2$
$\sigma_p^2 = (\omega_1\sigma_1 + \omega_2\sigma_2)^2$
$\sigma_p = \omega_1\sigma_1 + \omega_2\sigma_2$

(c) for $\rho = -1$
$\sigma_p^2 = (\omega_1\sigma_1 - \omega_2\sigma_2)^2$
$\sigma_p = |\omega_1\sigma_1 - \omega_2\sigma_2|$

So, for this to be 0,
$\omega_1\sigma_1 = \omega_2\sigma_2$
$\omega_1\sigma_1 = (1-\omega_1)\sigma_2$ $\quad (\omega_1 + \omega_2 = 1)$
$\omega_1(25) = (1-\omega_1)12$
$25\omega_1 = 12 - 12\omega_1$
$37\omega_1 = 12$

$$\boxed{\omega_1 = \frac{12}{37}}$$

So, for $\omega_1 = \frac{12}{37}$, risk becomes zero

5) I think its all about the diversification term we get that is $2\omega_1\omega_2\sigma_1\sigma_2\rho$. If it would not be there, then like expectation, Variance would also be the weighted average.
And when $\rho < 1$ which happens in most of the practical cases, $\sigma_p^2 < (\omega_1\sigma_1 + \omega_2\sigma_2)^2$, so its always below the weighted average of individual risks.
More importantly there is no sacrifice of the expected return, its still independent of correlation, thats why its called "free lunch".

6) (a) $\Sigma_{11} = 1(.25)(.25) = 0.0625$

$\Sigma_{22} = 1(0.12)(0.12) = 0.0144$

$\Sigma_{33} = 1(0.04)(0.04) = 0.0016$

$\Sigma_{12} = \Sigma_{21} = 0.4 \times 0.25 \times 0.12 = 0.0120$

$\Sigma_{13} = \Sigma_{31} = 0.1 \times 0.25 \times 0.04 = 0.0010$

$\Sigma_{23} = \Sigma_{32} = 0.2 \times 0.12 \times 0.04 = 0.00096$

$$\Sigma = \begin{bmatrix} 0.0625 & 0.0120 & 0.0010 \\ 0.0120 & 0.0144 & 0.00096 \\ 0.0010 & 0.00096 & 0.0016 \end{bmatrix}$$

(b) Since $\Sigma_{ij} = \Sigma_{ji}$ because of the commutative nature of the covariance that is $Cov(X,Y) = Cov(Y,X)$
Symmetry confirmed

7) (a) Sharpe ratio ($R_f = 4\%$)

$$S_p = \frac{E[R_p] - R_f}{\sigma_p}$$

$X \Rightarrow \quad S_x = \frac{14-4}{22} = 0.455$

$Y \Rightarrow \quad S_y = \frac{9-4}{10} = 0.5 \leftarrow$ highest

$Z \Rightarrow \quad S_z = \frac{20-4}{35} = \frac{16}{35} = 0.457$

(b) $Y > Z > X$

A rational investor would prefer portfolio Y as it has the highest sharpe ratio which indicates excess return per unit of risk.

(c) Though portfolio Z has the highest return (20%) but at the same time it has volatility of 35%. We are getting 11% extra return but 25% more risk when compared with Y.

So, a rational investor would look at return per unit of risk and would prefer Y.

8) (a) The global minimum variance (GMV) portfolio is the solution to the optimization problem.

$$\min_{w} \ w^T \Sigma w$$

$$\text{st} \qquad w^T \cdot 1 = 1$$

(no return target)

It is the leftmost point on the minimum variance frontier because it minimizes σ_p globally, that is portfolio with minimum risk

Every other portfolio has an additional constraint $w^T\mu = \mu^*$, which would have higher risk.

(b) No this is not correct, for every portfolio on the lower half, there exists a portfolio on the upper half which has the same (σ_p) risk but higher return. Regardless of risk tolerance one should always hold the portfolio on the upper-half.

(c) Capital market line (CML) is constructed by drawing a line from the risk free rate R_f (at $\sigma=0$) that is tangent to the efficient frontier.

The tangency point is market portfolio M.

Any point on the CML is achievable by mixing R_f and M in apporpiate ratio.

The slope of CML is sharpe ratio of M

$$S_M = \frac{E[R_M] - R_f}{\sigma_M}$$

9) (a) Standard error of mean

$$SE(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{n}} = \frac{0.3}{\sqrt{36}} = \frac{0.3}{6} = 5\%$$

(b) MVO allots highest weights on the assets with highest estimated $\hat{\mu}_i$. But those assets are likely to be upward biased means due to random noise. So, a small spike (noise) in $\hat{\mu}_i$ may lead to overeweighed allocation.

Whereas, ML predicted returns are less susceptible to this noise and can be considered stable.

(c) for $N = 50$

$$\text{unique entries} = \frac{N(N+1)}{2}$$

$$= \frac{50 \times 51}{2} = 1275 \text{ parameters}$$

Since we have only monthly data of 3 years, but 1275 parameters. The sample covariance matrix becomes unstable.
Small data changes can cause widely different optimal weights.
This is why estimation by ML is preferred.